

Micro Engraving Machine

Model: DK-8-KZ, DK-8 PRO-5, JZ-5, JZ-6, DK-5 PRO-5

User's Manual

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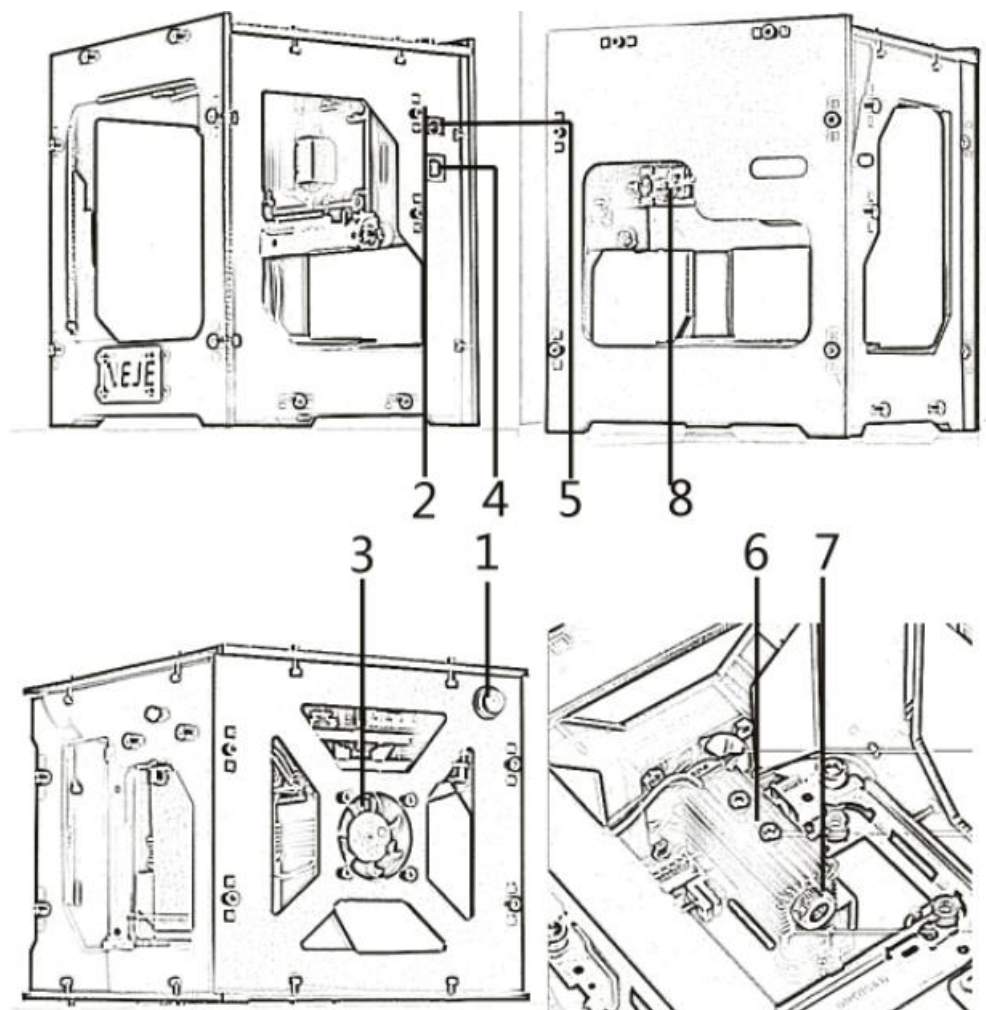
The laser engraver printer machine printer is a basic engraving machine, It is designed for desktop and laptop computers. The machine powered by a 5V1A power adapter usually. You can also change the power supply by a switch on the motherboard. and you can also connect it one by one, just use a computer, you can control several machines. You can also use a dual usb power adapter supply, and control it by the control buttons on the motherboard. These extensions provide more application scenarios for users.

Technical Parameter

Caving Space:	38 x 38 mm
Laser Diode:	Nichia 405nm
Communication Interface:	USB2.0
Image Size:	490x490 pixels(0.075x490=36.75mm) 350dpi
Supported image formats :	JPG , BMP (24 bit only) , TIFF , PNG (32 bit or less) , PCX (8 bit and 24 bit RLE) , PCD
Supported System:	Win Xp SP3\7\8\10
Engrave Mode:	Bitmap carving
Software Language:	English\German\Japanese\French\Italian\Chinese
Size:	Product size: 14.50 x 16.00 x 19.00 cm
	Package size: 30.00 x 20.00 x 21.00 cm
Every work time:	unlimited
	NOTE: Do not work long time in the high temperature environment
Laser can replace:	yes
Frame material:	Acrylic + Aluminum + Stainless Steel
Support Software:	NEJE Laser_Carver V3.5
Weight:	Product weight: 0.900 kg
	Package weight: 1.435 kg
Input Voltage:	5V 1A USB Power Adapter
Laser temperature:	25℃ - 45℃
Certificate:	CE,FCC,FDA
Objective engraving material:	"Smooth, Organic, opaque, Non- flame- retardant material For example: Wood, Gray cardboard, bamboo, Leather etc.
Unsuitable engraving material:	" Those material unable to burn &focus" For example: Pure metal(without coating), transparent plastic, White paper(Astigmatic),PP plastic etc.

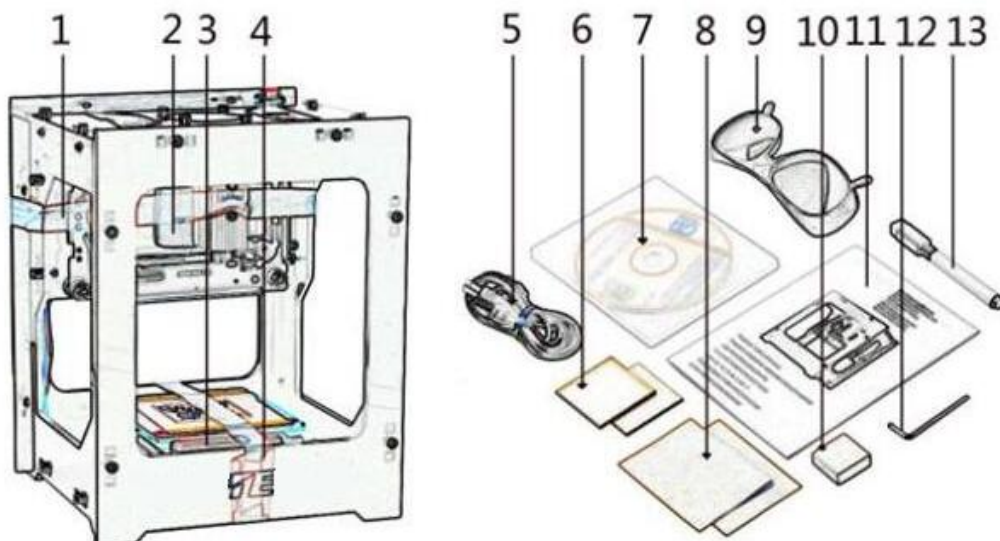
FEATURES

1	Multi-Function Button	Press 1: Box Positioning
		Press 2: Start
		Press 3: Pause
2	Signal Light	Red: Power
		Blue: Data
3	Cooling Fan	Cooling down the laser energy(heat)
4	USB Data Port	Transfer data
5	Power Port	5V1A Power Supply
6	Set Screws	Fix the laser stand on the radiator
7	Focusing Lens	Adjust the focal length
8	Offline control key	Control



Packing List

1 - 4	wrapping tape and pearl cotton	Remove all the tape and pearl cotton to start the machine
5	Data Calbe x 1	Power and transmit data
	Power Cable x 1	5V1A Power Supply
6	Wood slice x 2	Test sample
7	CD x 1	.net framework installer package, drive, software, image, etc.
8	gray cardboard x 2	Test sample
9	Glasses x 1	Support the focusing activity
10	Wood brick x 1	test sample
11	User manual x 1	Standard Operation instruction
12	L wrench x 1	For de-cap the machine cover to replace Laser head.
13	Whiteboard pen x 1	To black the surface of transparent material before engraving



How to use it with Windows Computers?

Prepare:

We will use the gray cardboard to carve as an example to explain, please follow the instructions to do your operation.

>> Obtain a desktop computer or laptop computer (computer operating system requirements is winXP, window 7, win8, win10);

>> Obtain a Usb rechargeable head (5V1A) (Apple, Samsung, Huawei mobile phone head can be, Support 5V1A or more);

>> Open the packaging, remove the fixed tape and pearl cotton on the machine;

>> Copy the CD or sd card files to the desktop, unzip the package to desktop.

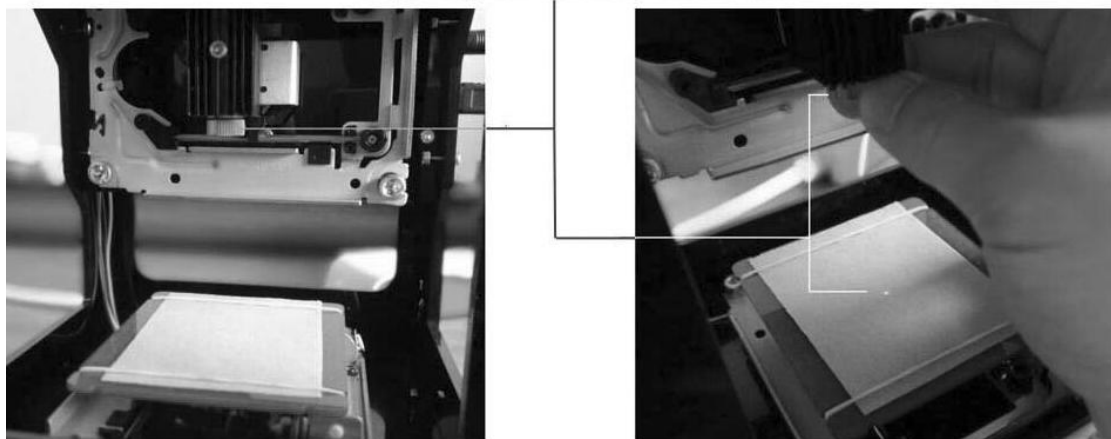
Or Simply download the application from the trusfer.com Web site, unzip the package, and follow the instructions contained in the Electronic tutorials on trusfer.com (recommend)

>> Adjust the focal length of the laser, factory default settings the focal length of the gray cardboard, there is no need to do any adjustment. If you need to carve the object in different thickness, you need to place the object in the engraving platform and adjust the focus manually, when in the first operation, it is recommended to use a gray cardboard (in the package list) to carve;

The method of adjust the focal length as follow:

1: Fixed the surface of the carved objects at the level of the bearing table, put on the focusing glasses(sent in random) (maybe will not adapt it at the beginning, just to be used as focus),rotate the focus head manually, shown in figure:

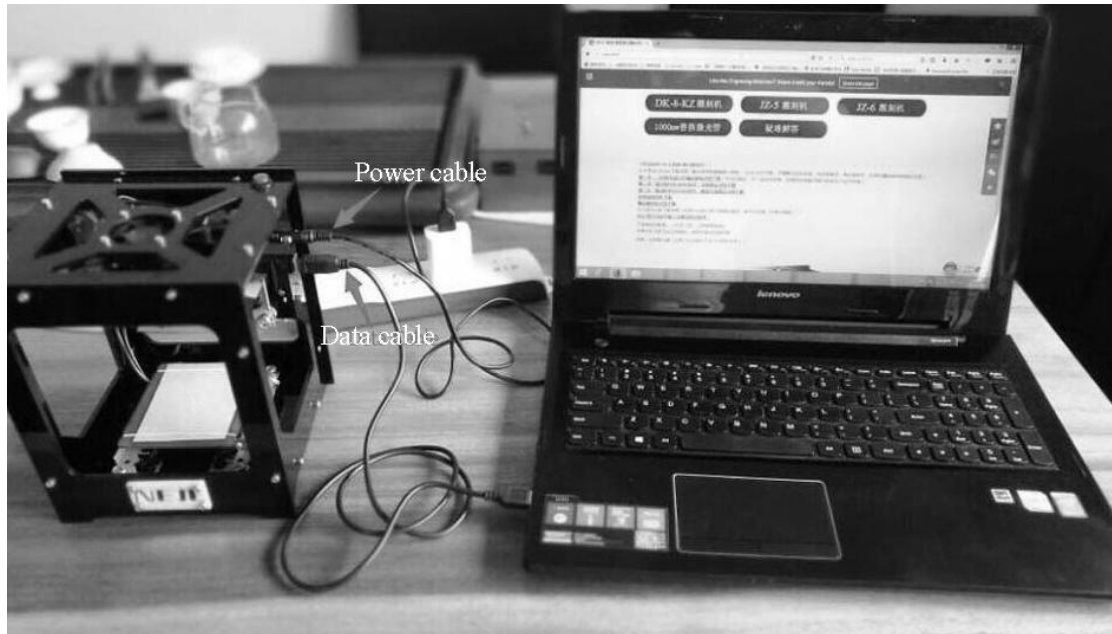
Adjusting the focuses of the lens



2: Observe the light down to the surface in the smallest point of the object , when the rotation to the smallest point, indicating that the focal length has been adjusted, note that the smaller the focus adjustment, the image engraving effect is more fine, if the focal length is not adjusted, there may be no image to be carve;

Step 1: Connection

As shown in the picture: Connect the round port usb cable to the 5V1A power adapter; connect the D port data cable to the computer usb port;



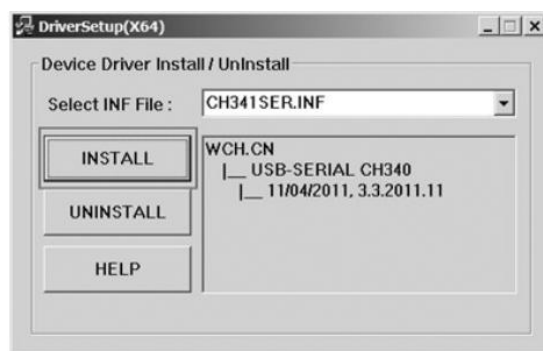
After the connection, the machine will detect the edge distance automatically , and then start the second step.

Step 2: Install the driver

it is recommended that you use the computer login URL: trusfer.com to download. download and decompress before the installation, first install the driver, then install the software, first connect the machine to the computer and then install.



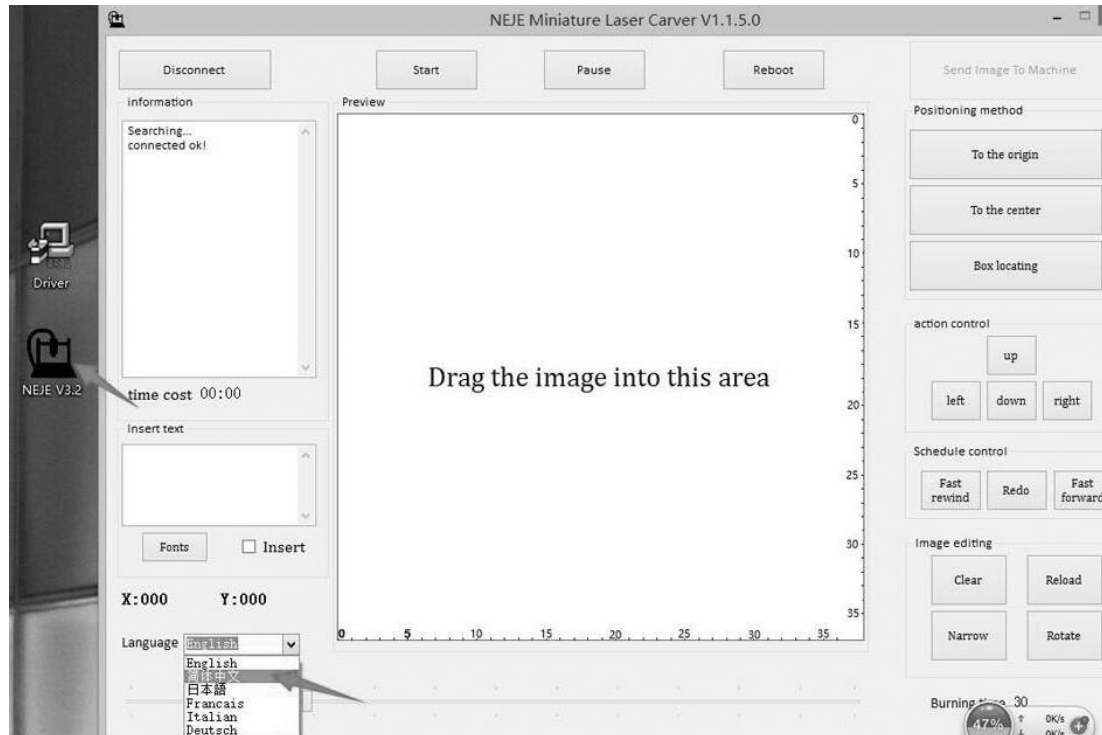
download and decompress, first install the driver, as shown below, after the success of the installation click to confirm bottom, and then close the installation window:



Step 3: Start the software

a. Open the software V3.5

The software will connect to the machine automatically, if you do not understand the English interface, please select Language!

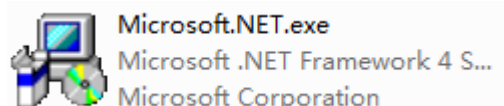


Note:

1. If you start the software, appear such a prompt (see below),



please install the .net framework 4.0.



System Requirements

Supported Operating System

Windows 7, Windows 7 Service Pack 1, Windows Server 2003 Service Pack 2, Windows Server 2008, Windows Server 2008 R2, Windows Server 2008 R2 SP1, Windows Vista Service Pack 1, Windows XP Service Pack 3

- Windows XP SP3
- Windows Server 2003 SP2
- Windows Vista SP1 or later
- Windows Server 2008 (not supported on Server Core Role)
- Windows 7
- Windows Server 2008 R2 (not supported on Server Core Role)
- Windows 7 SP1
- Windows Server 2008 R2 SP1

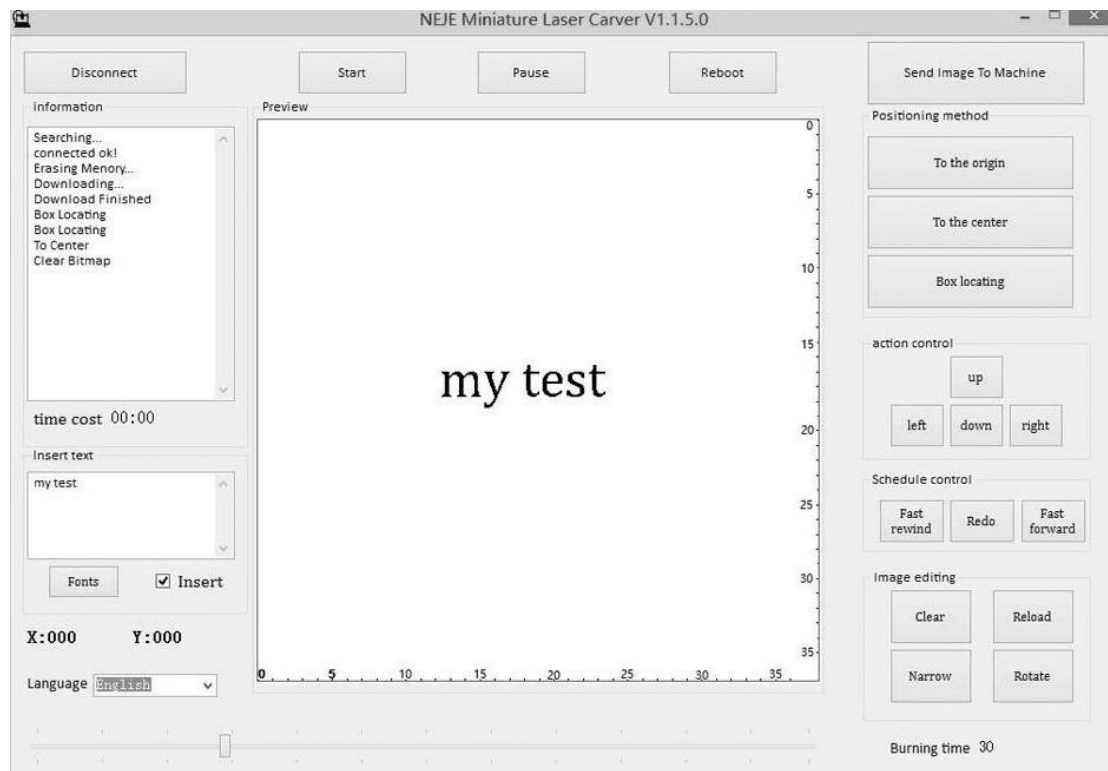
This is because the software is based on Microsoft .Net 4.0 platform development.

2. Win 8 and Win 10 are already installed, You don't have to install it. If your computer is XP SP1 or SP2 the .net 4.0 framework may not be able to support. If the packages with tf card or CD, Please copy to the desktop and extract it into desktop.

3. Do not Repeat open the software , otherwise the software will not be able to connect to the machine.

b. Test the text print

Enter "my test", you can select "font", select the font and the font size, and then click "insert", click the middle of the software area, then insert the text to print the area;

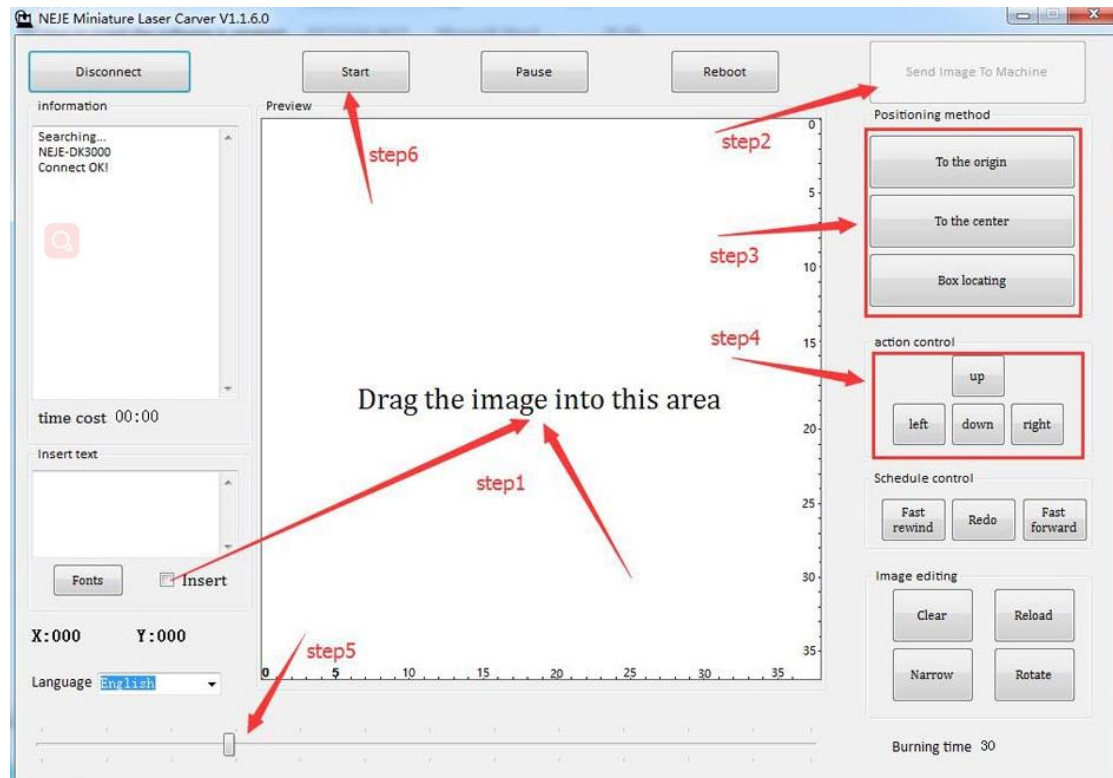


Then set the burn time (drag the slider, different materials, different burn depth should choose the different burn time, the test gray cardboard we choose the burn time 30) - click on "send pictures to engraving machine" (probably Wait for 5 seconds) - then click "Start".

Click on the start, the machine began to work.

also you can drag a jpg(490x490 px) picture to the area.

c.Setting and Start



As shown above, Please follow below steps to experience those function

Step1. Insert text, or drag into a JPG image of 550x550 pixels.

Step2. After edited, send pictures to the machine.

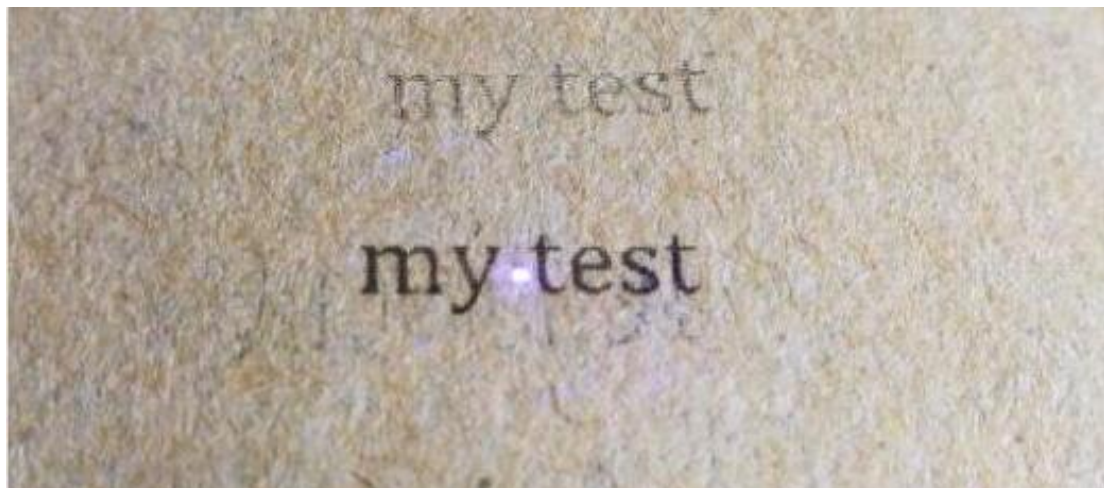
Step3. Positioning function: any point positioning, to the middle of the image, box positioning

Step4. Move the engraving position

Step5. Adjust the burning time, it is recommended to adjust the time about 40 for the first time using. Rotate the control to the left is to shorten the burning time, the shorter the time is, the lighter the carving depth is. vice versa.

Step.6 Start carving

Now, Let's start my first work.



Trouble shooting

1-- Unable to connect to Software how to deal with this situation?



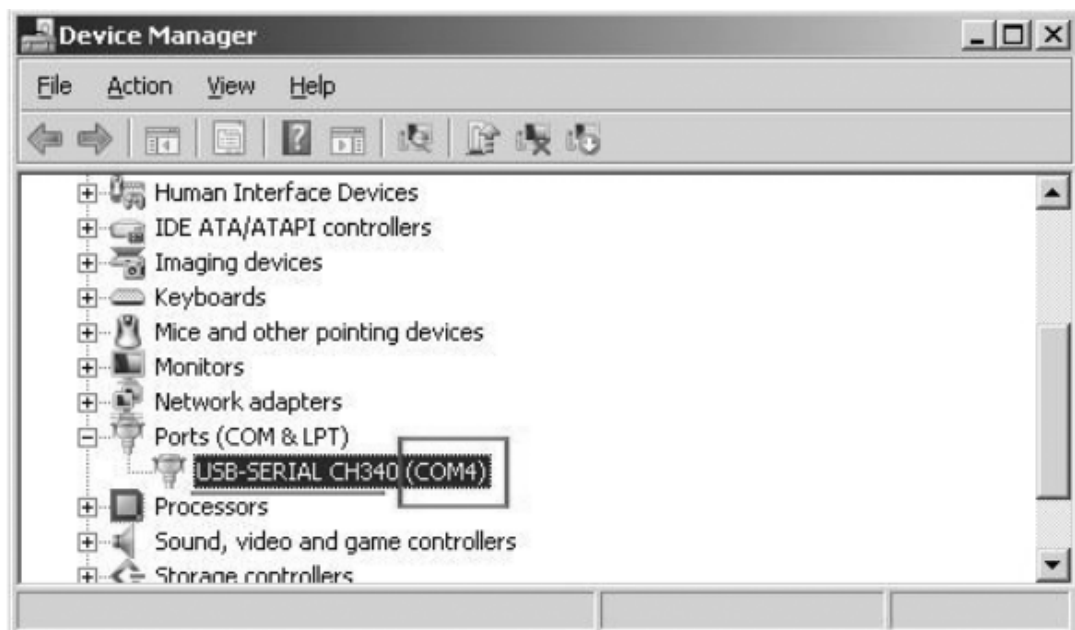
The software stays in wait for situation has 4 reasons:

A. Whether the drive is successfully installed

How to determine whether to install a drive successfully:

Plug the USB data line into the computer USB interface.

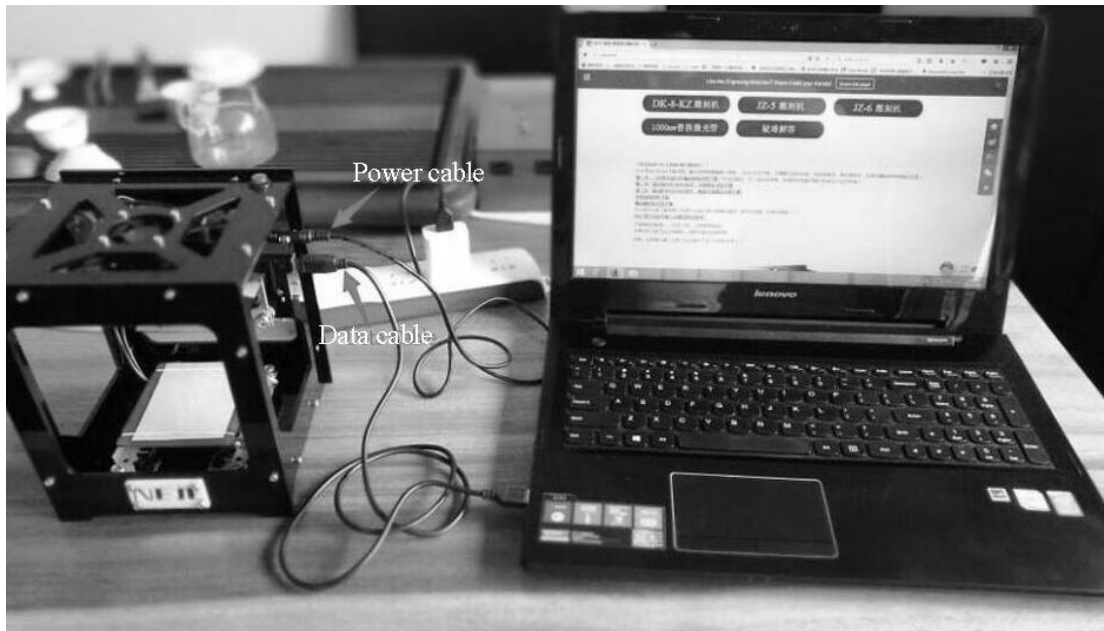
Click on the computer - my computer - Property - Device Manager - port, if you can find the CH340 port, indicating that the computer has identified the machine, drive installation successfully.



B. USB data line is out of function

B-1. Computer USB interface has bad contact. You can try to replace the computer USB interface, or replace the computer

B-2. Data line is defect. Replace a data line



C. Open the software repeatedly. If you open the software without closing it, then open the software again. There will be out of connection. In this case, turn off all the software and re-open the software.

D. You'd tried above all and still failed. Please turn off the machine and the computer again. This may be that the system has not yet accepted the new driver and needs to restart the computer.

2-- The machine works, but it doesn't carved out

Mainly root cause to below few aspects:

A. If the material is suitable for carving. Like the stone, pure metal, transparent, light, adding flame retardant materials of plastics are not suitable for carving. And if material is white, since there is some degree of astigmatism, figure carving effect might not be good either.

B. If the object is placed flat enough. If the surface of the object is uneven, or there're a lot of voids, this material is not suitable for engraving

C. If the object is well fixed. If the object has not been fixed by the rubber band or silica gel pad, it is very easy to make the engraving picture position deviated during engraving.

D. If the focus is well adjusted. The essence of the engraving machine is to focus the image by heat treatment, so you must well focus the laser to produce enough burning power.

E. If ignition timing is set properly. If the ignition time is too fast, it may cause un-complete burning in such a short time.

F. laser tube may burn out. There's life span for the laser tube, if it had been overload used, poor PM or even careless operation may finally lead to laser tubes burned.

3-- Why does the machine keep hitting a straight line?

The chassis is driven by a stepper motor. If the weight of the load is too heavy, the motor will not

be able to complete the drive. Please reduce the weight of the engraving object.

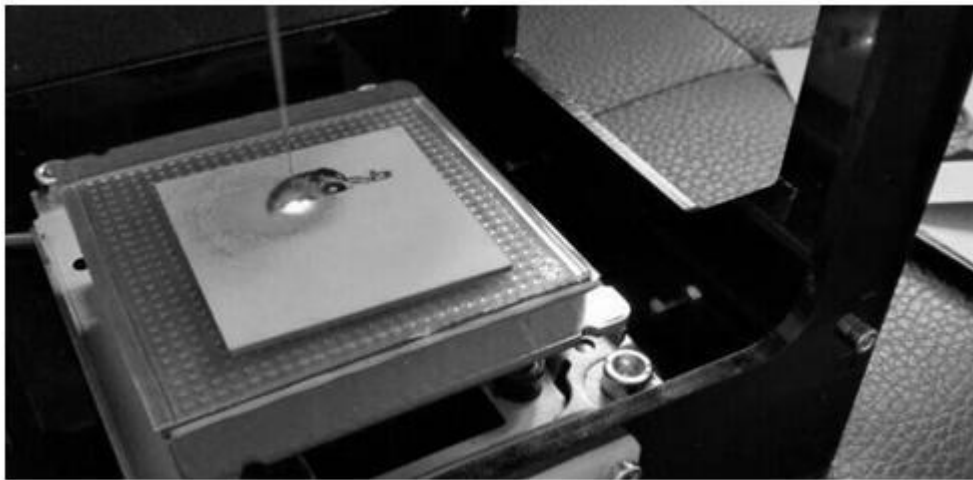
4--The machine driver is broken, what should I do?

The product parts can be found at the official website www.trusfer.com replacement, please make sure you have practical ability to replace it on your own. Or you should go to repair centre.

5--How to fix the engraved objects?

3options

1. Use the rubber band (sent in random);
2. Place the silicone pad on the bottom plate and place the object directly on the silicone pad (recommended) as shown below (the black silicone pad sent in random):



3. The user designs the card slot on his own.

6—How to do when it can not focus on?

There are 4 potential reasons listed as below:

- A. Object has been placed beyond the scope of the machine can focus. Normally, the laser tube focus range is around 5cm diameter on in the bottom plate .
- B. The object is transparent, unable to focus. In this case, you can use the whiteboard pen to turn the object surface to black to support focusing. Then wipe it up after curving.



- C. If the material is a little transparent, you can use gray cardboard to cover on the object's surface to support the focus adjustment.
- D. Laser tube light attenuation, poor spot may also lead to non focus

7--How many pictures can the machine store?

It can store only one picture that last sent to the machine;

8---How to choose engraving position?

Firstly, place the object to the middle of the bottom board.

The software provides 3 positioning tools:

10-1. Point positioning,

10-2. Position in the middle of the image,

10-3. Box positioning, can preview the engraving position in advance. Together with the direction key control, the control of any position can be realized.

9---How to set the depth of carving?

The depth of carving is determined by two factors:

1. Whether the focal length is fine;

2. Set the burning time (the longer time, the greater the depth, the slower the time)

10—what could I do if there's no power during engraving?

There's memory function in engraving machine. If power shortage, you can stop it, and continue the engraving from breakpoint after charging.

11---What to do if the focusing glasses is rickety and loosen?

The lenses are fixed by the spring with which during caving, there'll be enough power to support the stability lenses. And such level of rickety will not influent to the carving quality .And in order to ensure the focusing process can be relatively simple, focusing ring slot has been designed with specific tolerance and margin. If you accidentally drop the lenses, you can re-spring it in.

12--- If you want to have a high quality level of engraving effect?

The effect of engraving is determined by many comprehensive factors:

A. The properties of material itself

B. focal length

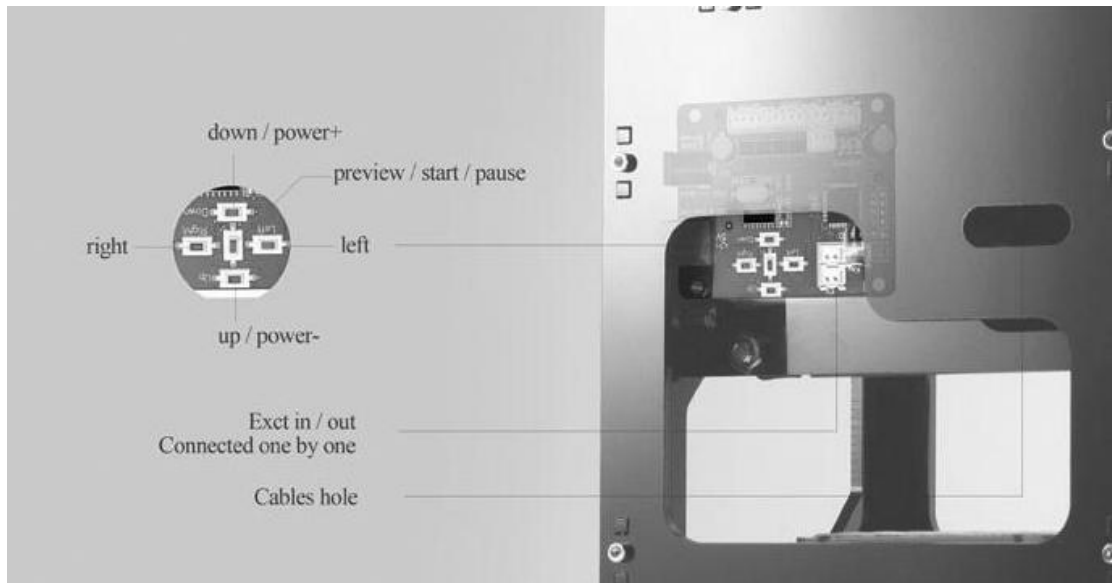
C. ignition time

D. If operate conform to the imaging principle: If the object has been placed flat enough, does it affect machine work?

etc.

13--How about offline engraving?

On the product back, there are left , right, position, start, pause and other multi-function bottom;; Insert two usb into the mobile power, control button can be make the offline engraving;



14--How about online engraving?

The motherboard back design 2 data interface, the need for parallel machine 2 usb offline energization, you can use 1 with 2pin 2.54 interface line, take the signal in the master machine, to achieve that the master machine drives multiple machines to work at the same time. One computer control N machines. (product list without the interface line accessories, you can DIY by yourselves if you interest in this function. Note, Interface line should be anti-connected)

15-- Carving half suddenly stopped, and self-acting?

This is because of the power is not enough, you can use a enough 5v1a usb power.

16--OEM?

phone: 86-18603021351

QQ: 867819731

Keep study, and master the knowledge slowly to complete the DIY, finally you can get what you want.

About laser head

Solid laser head

There's life span for the laser head , and laser type normally are CO2, Multimode high power laser, small solid-state laser, Our DK-BL uses Nichia blue 405nm single-mode laser diode, the official service life is 10000 hours, the normal use in about 3 months -1 years, but it's also closely related to how you use it .(main factors include : The temperature, time intensity, current, voltage, anti static treatment, adverse rate etc.)The laser module quality control technology

Laser module quality control process

The laser modules used by Our DK-BL is the best one that with constant current drive modules, along with a stable cooling system. And every single-mode module has being through 3 hours aging tested. and are well packaged before outgoing. So the machine that every user received had PASS a serial of reliability test. but even so, there may be 0.05% chance that the machine you

receive may have the laser durability problem. In this case, please confirm within 3 days, and feed back to the dealer, we can arrange replacement for you for free.

What about the laser head life?

There's service life for all laser products, the laser tube official life is 100,000 hours, but the actual life will depends on actual usage. In general, 3 Month - 1 year, or may be many years.

How to maintain a laser head?

To protect the laser' life, we suggest not use the machine more than 30 Minutes /time.

How to judge the laser head burned?

1. When the machine is working, the light was significantly weakened.
2. When the machine is working, the light scattered, unable to focus and the light weakened.
3. The burning time, Carving material, focal length all ok, but can't burn out the object.

Where to buy a new laser head for the machine?

Buy from the seller or buy from www.trusfer.com

How to replace the laser head?

Step1: use the L wrench Screw the screws on the back of the machine and open the back of panel.

Step2: use the L wrench Screw the screws on the radiator. and replace the laser tube.

Why the laser head focusing lens slightly twisted?

In order to ensure that the focusing lens and the optical port directly to the manual and flexible focus, and not designed very tight, but the middle of the support spring enough to support the focus lens in the course of the movement does not appear any shaking.



Caution

1. Our machine certified to CE\FCC\FDA certification, the radiation values far below the EU and the United States standards, You may rest assured to use

CE certification Safety not Have Radiation



2. Our machine laser energy is not that huge enough to damage you if hand or other part of body stay in the curving scope accidentally during curving, but, you'd better be careful since it's abnormal operation of machine. We can't assure it's no harm definitely.
3. Do not take out the laser head, and exposure to body!



4. Do not use in where the children can touch!
5. Keep away from flammable and explosive materials!
6. There's an emergency pause button designed on the top of machine that can be used in an emergency!

Maintenance

- To ensure that the motor drive clean.
- If you don't use it for a long time, wrap it up as best as possible and avoid dust affecting machine performance.
- Put it in a safe place to avoid dropping.
- ESD-proof for the machine to prevent ESD destroy the laser.



Engraving machine is a kind of electrical product, should be operated in a standard and safe manner to prevent safety issue occur.

For your safety concern, Please read this User Manual carefully before operating. And always keep this Manual ready.

You can also login in to www.trusfer.com via your computer or Scan below code via phone to read the E-Manual .

Manufacturer: Shenzhen Zhi Xin Jie Technology Co., Ltd | Tel:86-186 0302 1351 | QQ: 867819731 | CE / FDA